

Final Assignment

Lung Cancer Prediction Using Machine Learning Classification Models

Name: Pranish Mulmi

University id: 2462261

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Abstract

One of the major causes of cancer related mortality globally is lung cancer whose early detection and diagnosis is of paramount importance in enhancing the survival of the patients. As more and more medical data becomes accessible, machine learning methods provide a viable solution to predict the outcomes of lung cancer, using the characteristics of the patient and the medical indicators.

This project aims at creating machine learning classification models that can be used to predict the existence of lung cancer using a structured lung cancer prediction dataset. It is an end-to-end machine learning study, which involves exploratory data analysis, data preprocessing, model development, hyperparameter optimization, and feature selection. The classical machine learning models and a neural network classifier were developed and tested with the help of a fitting classification measure. This piece of work is in line with the United Nations Sustainable Development Goal 3 (Good Health and Well-being) as it promotes making healthcare decisions that are based on data.

Introduction

Problem Statement

The diagnosis of lung cancer usually depends on the medical tests, imaging, and clinical judgment of the specialists. Manual diagnosis, however, might be time consuming and postpone treatment to a system that is resource limited. Computational models can help healthcare professionals by allowing the early prediction of the presence of lung cancer offering quick and credible decision support.

The prediction of lung cancer in this project is developed as a two-class problem of classification, i.e., one wants to categorize the patients with cancer-positive or cancer-negative patients due to the clinical and lifestyle-related factors.

Dataset Description

The datasets that were used to predict lung cancer are attributes of patients which include demographics characteristics, lifestyle choices and medical conditions known to be associated with lung cancers. The data has several predictors in the form of features, and a dependent variable which has the presence or absence of lung cancer.

The data set is a realistic model of healthcare situation and offers an appropriate foundation to implement supervised classification algorithms. The project can be applied to UNSDG 3 because predictive analytics in healthcare can enhance patient outcomes and early diagnosis.

Objectives

The key things the classification task will focus on include:

- To investigate and learn about the lung cancer dataset with the help of exploratory data analysis.
- To pre-process and pre-treat the data to be classified into models.
- To adopt classical and neural network models of classification.
- To optimize the model performance with hyperparameter tuning.
- In order to implement feature selection methods.
- To compare the performance of the models with the right evaluation metrics.
- Only a handful of studies utilize it, yet it remains an essential component of the research process.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to learn more about the data and see the distributions of features and what problems may occur with them, e.g. imbalanced classes and outliers.

Figure 1: Distribution of the targets class.

Target class distribution plot demonstrates the patients who have and do not have lung cancer. Such a visualization can be used to evaluate the imbalance in classes, of which this is also a significant factor in choosing evaluation metrics.

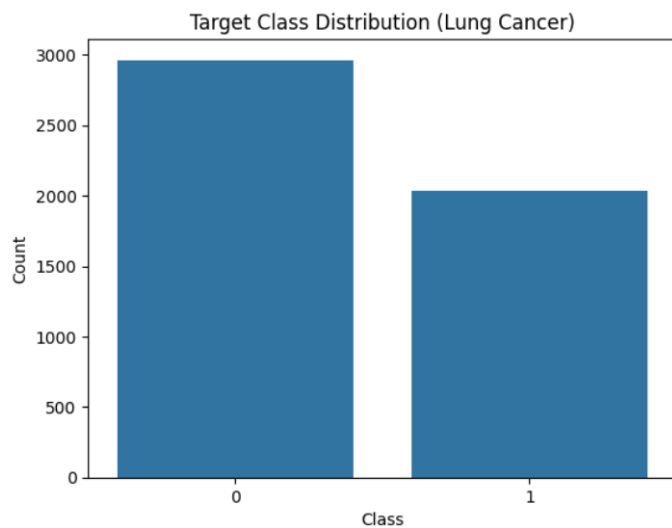
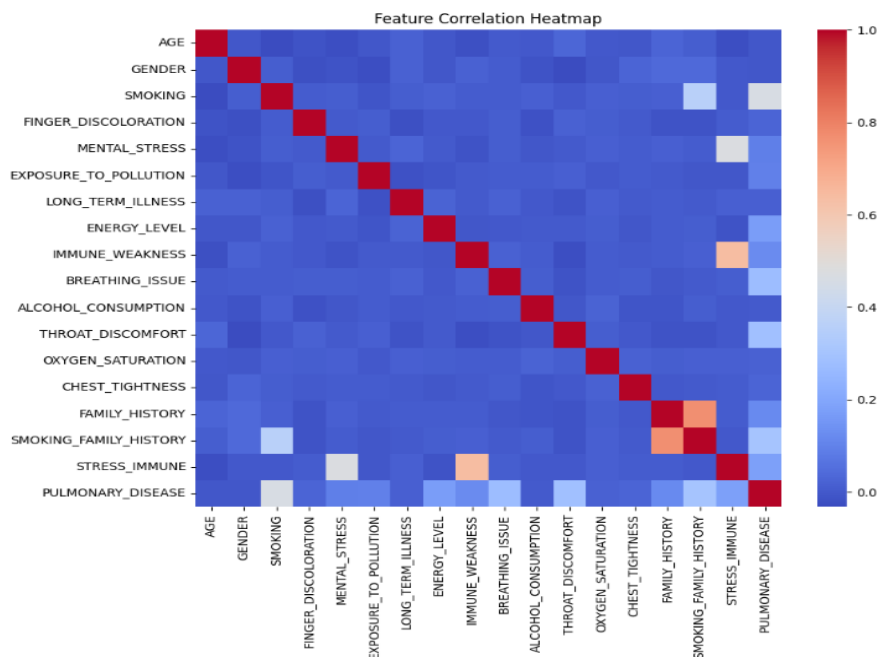


Figure 2: Correlation Heatmap

The correlation heatmap shows the associations between numerical features and potentials of multi-collinearity. The knowledge of these relationships facilitates the feature selection and interpretation of the model.



Methodology

Data Preprocessing:

Preprocessing of data was done by transforming categorical variables into numerical forms that could be used in machine learning algorithms. Being analysed prior to modelling, the dataset was tested on the absence of values and inconsistency. Numerical features were made more standardized through feature scaling, which is especially significant to the distance-based models and neural networks.

Stratified sampling was used in subdivision of the dataset into training and testing sets to maintain the original class distribution and provided an unbiased assessment.

Model Building:

Neural Network Classifier

The implementation of a neural network classifier is a Multi-Layer Perceptron (MLP) architecture that is applied in order to obtain complex and non-linear associations between the input features and the target variable. The model consists of the hidden layers with non-linear activation functions, and was trained as a result of an iterative optimization.

Classical Classification Models.

Besides the neural net, two standard machine learning classifiers were developed:

Logistic Regression that gives a simple and interpretable baseline model.

K-Nearest Neighbors (KNN), that groups samples according to the similarity in the feature space.

The approaches to be compared using these models are linear, distance-based and neural network approaches.

Model Evaluation:

These metrics were used to estimate the performance of classification models:

- Accuracy
- Precision
- Recall
- F1-score

The F1-score was highlighted because it averages precision and recall and is hence very appropriate when dealing with medical classification where false positive and false negative are very crucial.

Figure 3: Confusion Matrix

The confusion matrix offers a break in of correct and incorrect predictions in detail and demonstrates true positives, true negatives, false positives, and false negatives.

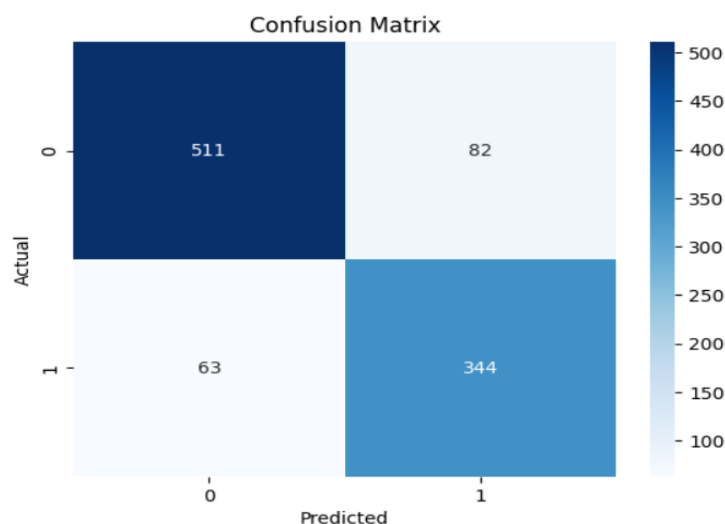
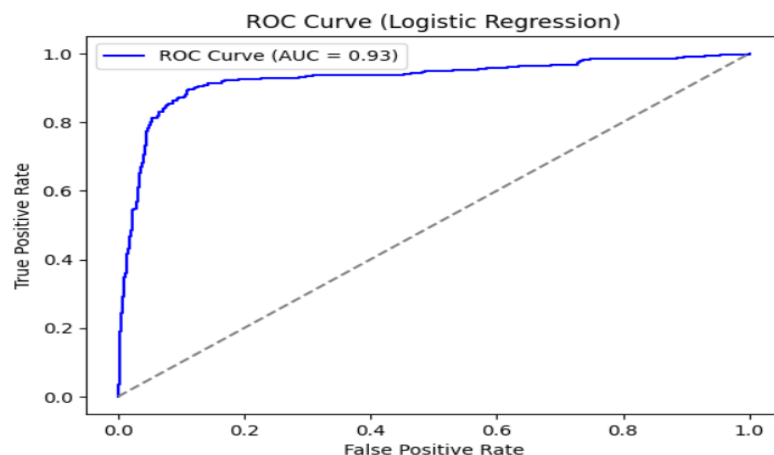


Figure 4: ROC Curve

ROC curve was utilized to test the capability of the model to differentiate between classes at varying thresholds.



Hyperparameter Optimization:

The cross-validation techniques were applied to achieve hyperparameter tuning to enhance the overall generalization of the model and minimize overfitting. The grid search was used to determine the best parameter combinations to be used in best models leading to higher classification performance.

Feature Selection:

The features related to the prediction of lung cancer were categorized as the most relevant ones using the techniques of feature selection. In a subset of informative features is useful in reducing model complexity, enhancing interpretability, and increasing predictive performance. The chosen features were used to retrain the models and measure performance improvements.

Result & Conclusion

Model	CV Score	Accuracy	Precision	Recall	F1-Score
Neural Network	—	0.874	0.863049	0.820639	0.841310
Logistic Regression	0.877056	0.887	0.848341	0.879607	0.863691
KNN	0.840018	0.851	0.804245	0.837838	0.820698

This project showed the use of machine learning classification in predicting lung cancer. Reliable predictive models were created using exploratory data analysis, model development, hyperparameter optimization, and feature selection.

The findings offer support to the possibility of machine learning to assist in early cancer detection and clinical decision-making. Additional clinical data, ensemble learning, and validation of models on external datasets may be considered as the elements of future work to increase the reliability and robustness of the model.

Discussion

- Model Performance: The Logistic Regression had a constant accuracy and F1-score.
- Effects of Hyperparameter Optimization: better results of KNN and Logistic Regression.
- Feature Selection: Dimensionality reduction and better interpretability.
- Limitations: The size of the dataset was relatively small and could not be generalized.
- Future Studies: Examine ensemble algorithms (Random Forest, Gradient Boosting), increase the scale of the dataset, and use sophisticated feature engineering.

References

- Kaggle: Lung Cancer Prediction Dataset (TheDevastator, 2021).
- Link: <https://www.kaggle.com/datasets/shantanugarg274/lung-cancer-prediction-dataset>
- Libraries: pandas, numerpy, matplotlib, seaborn, scikit-learn.
- UN Sustainable Development Goals: Goal 3 - Good Health and Well-Being.